TECHNICAL MONOGRAPH SERIES IN ARTIFICIAL INTELLIGENCE • VOL. 1

OmniDoc-RAG: Visual Document Retrieval, 2D Spatial Rotary Embeddings, and Multimodal Late-Interaction Reasoning

An OCR-Free Deep Learning Framework for Dense Document Retrieval, Spatial Layout Preservation, and Grounded Question Answering

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Open Source Research Monograph • August 2026

Repository: `github.com/nizamulhaq500/OmniDoc-RAG`

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FRONT MATTER

Executive Abstract

Modern enterprise document architectures rely overwhelmingly on Optical Character Recognition (OCR) heuristics or rule-based string parsers to ingest multi-page PDF documents into text-only Retrieval-Augmented Generation (RAG) pipelines. While effective for unformatted narrative text, this linear serialization paradigm collapses catastrophically when applied to complex visual layouts—including hierarchical financial statements, multi-column scientific publications, engineering blueprints, and dense tabular ledgers. By converting two-dimensional visual topologies into one-dimensional token streams, spatial coordinate structures, bounding geometry, reading flows, and non-textual graphical primitives are permanently annihilated.

This monograph establishes **OmniDoc-RAG**, an end-to-end, OCR-free multimodal document retrieval and question-answering architecture that operates directly on raw high-resolution document image manifolds ($\mathbb{R}^{3\times1024\times1024}$). The technical contributions articulated in this document encompass:

- **2D Spatial Rotary Position Embeddings (2D-RoPE):** We formulate and prove the spatial relative coordinate translation invariance of an orthogonal frequency decomposition that preserves 2D Euclidean document geometry under multi-head attention.
- **Perceiver Resampler Latent Bottleneck:** We design a cross-attention compression module that reduces $1,024$ spatial visual patch tokens down to 64 latent representations ($16\times$ compression), eliminating the quadratic attention barrier of multi-page visual inference.
- **Multi-Scale Symmetric Patch-InfoNCE Criterion:** We derive the analytical loss gradients and convergence bounds of a late-interaction (MaxSim) contrastive objective equipped with a learnable log-inverse temperature parameter τ, optimized across $39,463$ real-world document QA pairs over $12,330$ steps on NVIDIA Blackwell hardware.
- **Grounded Generative Reader:** We integrate retrieved visual latents into a local Vision-Language Model (Qwen2.5-VL) via Low-Rank Adaptation (LoRA), enforcing masked causal cross-entropy loss to prevent hallucination.

On an unconstrained $1,000$-document real-world evaluation benchmark, OmniDoc-RAG demonstrated a $10.2\times$ **surge in Mean Reciprocal Rank (MRR)** (advancing from 0.0172 baseline to 0.1746) and achieved a **Recall@5 of** 27.50%, while driving training loss down from 4.3742 to 0.3793 (a -91.3% drop). All software implementations are validated by a formal 32-unit test suite executing with 100% pass rate.

PART I

System Architecture, Engineering & Empirical Findings

CHAPTER 1

The Breakdown of Text-Based OCR RAG

1.1 The Classical Serialization Pipeline

Conventional document retrieval pipelines construct vector representations by stripping formatting through an OCR engine (e.g., Tesseract, EasyOCR) or a PDF text scraper (e.g., PyPDF, PDFMiner). The resulting character stream is partitioned into fixed-length chunk windows (typically 256–512 tokens), embedded via a dense text encoder (such as `text-embedding-3` or `bge-large`), and queried using single-vector cosine similarity.

1.2 Taxonomy of Systemic Failure Modes

Failure Mode 1 (Reading Order Collapse). In multi-column publications, bounding-box line grouping algorithms frequently interleave sentences horizontally across column boundaries, yielding nonsensical context blocks that corrupt downstream language generation.

Failure Mode 2 (Tabular Coordinate Destruction). Spreadsheets and financial balance sheets encode critical semantic relationships along vertical columns and horizontal rows. Flattening a matrix into a delimited text string destroys row-header and column-header coordinate associations.

Failure Mode 3 (Graphic & Schematic Blindness). Vector plots, flowcharts, chemical diagrams, signature approvals, and visual callout boxes possess zero textual tokens and are discarded from the retrieval index entirely.

Failure Mode 4 (OCR Character Noise & Font Artifacts). Specialized mathematical fonts (∇, $\int$, $\sum$), rotated typography, and watermark backgrounds trigger severe Character Error Rates (CER), polluting embeddings with out-of-vocabulary artifacts.

CHAPTER 2

The OmniDoc-RAG Vision-Native Paradigm

2.1 High-Resolution Visual Ingestion

OmniDoc-RAG eliminates text-extraction heuristics by modeling each document page as a continuous image manifold $\mathbf{I} \in \mathbb{R}^{3\times1024\times1024}$ rendered at $150\ \mathrm{DPI}$ with isotropic aspect-ratio scaling and symmetric white padding. A $2\mathrm{D}$ convolutional patch projection with kernel size $P = 32$ and stride $S = 32$ extracts $N = (1024/32)^2 = 1,024$ spatial patch tokens of hidden dimension $D = 768$.

2.2 Two-Stage Coordinated Retrieval & Generation

The OmniDoc-RAG architecture executes two coordinated stages:

1. **Stage 1 (Dense Visual-Semantic Retrieval):** Evaluates multi-vector late-interaction similarity (MaxSim) between query subwords and the 64 latent representations of candidate document pages, executing millisecond top-K candidate retrieval.
2. **Stage 2 (Grounded Visual Question Answering):** Feeds the retrieved visual latents directly into the causal attention mechanism of a local Vision-Language Model (VLM) via Low-Rank Adaptation (LoRA), synthesizing factual answers grounded strictly in the visual page.

CHAPTER 3

Codebase Architecture & Module Specifications

The OmniDoc-RAG software repository is structured modularly. The table below delineates the file-to-logic mapping across all functional components:

Module Path	Primary Classes / Functions	Architectural Responsibility
`models/rope2d.py`	`RotaryEmbedding2D`, `rotate_half`	Continuous 2D spatial rotary frequency generation, manifold duplication, and key rotation.
`models/perceiver.py`	`PerceiverResampler`	Multi-head cross-attention bottleneck compressing $1,024$ patches to 64 latents.
`losses/contrastive_loss.py`	`SymmetricPatchInfoNCELoss`	Dual-directional contrastive loss with learnable inverse temperature and MaxSim scoring.
`models/scaled_omni_encoder.py`	`ScaledOmniDocDualEncoder`	Production dual-encoder pairing `bert-base-uncased` with 2D-RoPE Perceiver vision backbone.
`data/pdf_processor.py`	`PDFProcessor`	PyMuPDF 150 DPI rendering, isotropic scaling, white margin padding, and ImageNet normalization.

Module Path	Primary Classes / Functions	Architectural Responsibility
`data/docvqa_dataset.py`	`DocVQADataset` , `FastDocVQADataset`	Zero-RAM memory-mapped Apache Arrow streaming and pre-tokenized GPU question buffers.
`train_scale_pretraining.py`	`train_scaled()`	Cloud pretraining engine with FP16 automatic mixed precision and gradient accumulation.
`train_stage2_vlm.py`	`Stage2VLMAdapter`	Generative reader training connecting visual latents to Qwen2.5-VL via LoRA ($r = 16$).
`evaluate_benchmarks.py`	`evaluate_retrieval()`	Evaluation harness calculating Recall@1, Recall@5, Recall@10, and Mean Reciprocal Rank (MRR).
`app.py`	`compute_semantic_retrieval()`	Interactive Streamlit app with sliding-window paragraph max-pooling and neural sentence QA.
`tests/test_*.py`	32 Pytest Verification Units	Full unit test suite guaranteeing mathematical invariants, shapes, and pipeline determinism.

CHAPTER 4

Systems Engineering Breakthroughs & Diagnostics

4.1 Diagnostic Post-Mortems

Breakthrough 1 (2D-RoPE Shape Broadcasting Invariant). Slicing attention head dimension $D_h = 64$ across two spatial axes produced vertical and horizontal frequency blocks of dimension 16. Concatenating y and x yielded a frequency manifold of shape $(1024, 32)$, triggering dimensional mismatch against query tensors of shape $(B, H, 1024, 64)$. Resolved by constructing an exact manifold duplication $[\mathbf{\Theta}_{2D}, \mathbf{\Theta}_{2D}] \in \mathbb{R}^{1024\times64}$, preserving rotational invariance while enabling vector broadcasting.

Breakthrough 2 (Zero-RAM Apache Arrow Disk Indexing). Ingesting $10, 194$ document pages as uncompressed PIL Python objects consumed > 440 GB of virtual memory, triggering operating system SIGKILL events. Resolved by engineering `LazyFullDocVQADataset` on top of HuggingFace Arrow memory-mapped binary files. Pages are decompressed directly into uint8 PyTorch tensors on-demand in the active batch slice, reducing host RAM overhead to < 1 MB.

Breakthrough 3 (Resolving CPU Data Starvation on Blackwell Accelerators). In initial cloud pretraining runs, the single-thread CPU execution of PIL Lanczos image resampling and subword tokenization took 2.8s per batch of 32, starving the NVIDIA Blackwell GPU (utilization $< 5\%$). Implemented a Turbo Data Pipeline: pre-tokenized all $39, 463$ questions into a persistent GPU tensor buffer ($O(1)$ lookup), replaced Lanczos filtering with fast bilinear scaling, and offloaded ImageNet floating-point normalization directly onto CUDA tensor cores. Per-step latency dropped from 3.0s to 0.08s (> 12 steps/sec), achieving a $> 10\times$ pretraining acceleration.

Breakthrough 4 (Query Intent Boundary Resolution in Production Serving). Punctuation attachments in raw queries (e.g. `"what is a codec?"` retaining `"codec?"`) broke token-matching filters and caused global page mean-pooling to dilute short, high-value definition sentences. Resolved by deploying regex-based keyword parsing (

> \textbackslash bkeyword\textbackslash b), multi-paragraph sliding window max-pooling, and BERT contextual sentence ranking.

CHAPTER 5

Empirical Convergence & Quantitative Benchmarks

5.1 Pretraining Convergence Metrics (12,330 Steps)

OmniDoc-RAG underwent full-scale contrastive pretraining across 10 epochs comprising $12,330$ optimization steps on the complete $10,194$-page `vikhyatk/docvqa` dataset ($39,463$ document-query pairs). Training executed with an effective batch size of 32 (16 per forward pass with gradient accumulation steps 2) and FP16 automatic mixed precision.

Pretraining Stage	Steps	Hardware	Start Loss	End Loss	Loss Delta	Convergence Phase
Stage 1 Sanity	186	Apple M2 Pro (CPU)	2.6535	2.1300	-19.7%	Initial gradient flow check on 1k samples.
Scaled 5-Epoch	155	Cloud GPU	2.7809	2.0885	-24.9%	BERT subword encoder integration.
Scaled 10-Epoch	310	Cloud GPU	7.2140	2.0798	-71.2%	Plateaus near batch entropy bound $\ln(32) \approx 3.46$.
Full Pretraining (Final)	**12,330**	**NVIDIA RTX PRO 6000 Blackwell**	**4.3742**	**0.3793**	**-91.3%**	**Deep visual-language cross-modal alignment across 39.5k QA pairs.**

5.2 Retrieval Performance vs Baseline Methods

Retrieval benchmarks were evaluated on an unconstrained test collection of $1,000$ real document pages from DocVQA using standard information retrieval metrics:

Retrieval Method	Pretraining Scale	Test Size	Recall@1	Recall@5	Recall@10	MRR	Relative MRR Gain
Random Uniform Selection	None (0 steps)	1,000 docs	0.10%	0.50%	1.00%	0.0074	Baseline $(1.0\times)$
Stage 1 Prototype	186 steps (1k docs)	1,000 docs	0.50%	1.30%	2.80%	0.0172	$+132\%$ ($2.3\times$)
Scaled Dual-Encoder	155 steps (1k docs)	1,000 docs	0.50%	2.50%	4.80%	0.0294	$+297\%$ ($4.0\times$)
Scaled Dual-Encoder	310 steps (1k docs)	1,000 docs	0.50%	3.50%	6.20%	0.0302	$+308\%$ ($4.1\times$)
OmniDoc-RAG (Final)	**12,330 steps (39.5k QA)**	**1,000 docs**	**6.40%**	**27.50%**	**38.90%**	**0.1746**	**+2,259% (23.6× vs Random / 10.2× vs Stage 1)**

PART II

Comprehensive Mathematical Foundations & Formal Proofs

CHAPTER 8

Continuous 2D Metric Spaces & Patch Geometry

8.1 The Continuous Page Surface

Let a document page be defined on a compact continuous 2D Euclidean metric manifold $\mathcal{S} = [0, H] \times [0, W] \subset \mathbb{R}^2$. Let $\mathbf{I} : \mathcal{S} \to [0, 1]^3$ denote the continuous vector-valued color field.

8.2 Patch Discretization & Projection

Under non-overlapping square patch partitioning with patch side length $P = 32$:

$$\Phi : \mathbb{R}^{3 \times H \times W} \to \mathbb{R}^{N \times (3P^2)}, \quad N = \frac{H}{P} \times \frac{W}{P}$$

A linear convolution projection transforms each spatial receptive field into hidden dimension $D = 768$:

$$\mathbf{X}_{i,j} = \sum_{c=1}^{3} \sum_{u=0}^{P-1} \sum_{v=0}^{P-1} \mathbf{W}_{\text{proj}}[d, c, u, v] \cdot \mathbf{I}[c, iP + u, jP + v] + \mathbf{b}_{\text{proj}}[d]$$

CHAPTER 9

Mathematical Derivation of 1D RoPE

9.1 Relative Position Inner-Product Constraint

Given query $\mathbf{q}_m$ at sequence index m and key $\mathbf{k}_n$ at index n, we require an encoding function ϕ such that the attention inner product depends strictly on relative position $m - n$:

$$\langle \phi(\mathbf{q}_m, m), \phi(\mathbf{k}_n, n) \rangle = g(\mathbf{q}_m, \mathbf{k}_n, m - n)$$

9.2 The Complex Exponential Solution

In a 2D subspace isomorphic to $\mathbb{C}$, vector $(x_1, x_2)^\top$ corresponds to $z = x_1 + ix_2$. Rotational transformation by angle $m\theta$ yields:

$$\mathbf{R}_m(\theta) = \begin{pmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{pmatrix} \in \mathrm{SO}(2)$$

Because $\mathbf{R}_m(\theta)^\top \mathbf{R}_n(\theta) = \mathbf{R}_{n-m}(\theta)$, the inner product satisfies:

$$\langle \mathbf{R}_m(\theta)\mathbf{q}, \mathbf{R}_n(\theta)\mathbf{k} \rangle = \mathbf{q}^\top \mathbf{R}_{n-m}(\theta)\mathbf{k} = g(\mathbf{q}, \mathbf{k}, n - m)$$

CHAPTER 10

2D Spatial Rotary Embeddings (2D-RoPE)

10.1 Direct Sum Decomposition

Let attention head dimension be $D_h = 64$. We decompose $\mathbb{R}^{D_h}$ into two orthogonal direct subspaces corresponding to vertical row coordinate y and horizontal column coordinate x:

$$\mathbb{R}^{D_h} = \mathcal{V}_y \oplus \mathcal{V}_x, \quad \dim(\mathcal{V}_y) = \dim(\mathcal{V}_x) = \frac{D_h}{2} = 32$$

Each subspace is structured into 16 mutually orthogonal 2D rotational planes with geometric frequency basis:

$$\theta_k = 10000^{-k/16}, \quad k \in \{0, 1, \dots, 15\}$$

Theorem 1 (2D Spatial Translation Invariance). *For any query $\mathbf{q} \in \mathbb{R}^{D_h}$ at spatial coordinate $\mathbf{p}_1 = (y_1, x_1)$ and key $\mathbf{k} \in \mathbb{R}^{D_h}$ at spatial coordinate $\mathbf{p}_2 = (y_2, x_2)$, the inner product under $\mathbf{R}_{2D}$ depends strictly on the spatial displacement vector $\Delta\mathbf{p} = \mathbf{p}_1 - \mathbf{p}_2$:*

$$\langle \mathbf{R}_{2D}(\mathbf{p}_1)\mathbf{q}, \mathbf{R}_{2D}(\mathbf{p}_2)\mathbf{k} \rangle = g(\mathbf{q}, \mathbf{k}, y_1 - y_2, x_1 - x_2) = g(\mathbf{q}, \mathbf{k}, \Delta\mathbf{p})$$

Proof. Decompose $\mathbf{q} = \mathbf{q}_y \oplus \mathbf{q}_x$ and $\mathbf{k} = \mathbf{k}_y \oplus \mathbf{k}_x$. By block-orthogonality:

$$\mathbf{R}_{2D}(\mathbf{p}_1)^\top \mathbf{R}_{2D}(\mathbf{p}_2) = \operatorname{diag}\left(\mathbf{R}_{y_1}^\top \mathbf{R}_{y_2},\ \mathbf{R}_{x_1}^\top \mathbf{R}_{x_2}\right)$$

Applying the 1D identity to each block:

$$\mathbf{R}(y_1\theta_k)^\top \mathbf{R}(y_2\theta_k) = \mathbf{R}((y_2 - y_1)\theta_k)$$
$$\mathbf{R}(x_1\theta_k)^\top \mathbf{R}(x_2\theta_k) = \mathbf{R}((x_2 - x_1)\theta_k)$$

Summing across all orthogonal planes:

$$\langle \mathbf{R}_{2D}(\mathbf{p}_1)\mathbf{q}, \mathbf{R}_{2D}(\mathbf{p}_2)\mathbf{k} \rangle = \sum_{k=0}^{15} \mathbf{q}_{y,k}^\top \mathbf{R}((y_2 - y_1)\theta_k)\mathbf{k}_{y,k} + \sum_{k=0}^{15} \mathbf{q}_{x,k}^\top \mathbf{R}((x_2 - x_1)\theta_k)\mathbf{k}_{x,k}$$
$$= g(\mathbf{q}, \mathbf{k}, \Delta\mathbf{p}) \quad \blacksquare$$

CHAPTER 11

Perceiver Resampler Latent Bottleneck

11.1 Mathematical Formulation

Let $\mathbf{Z} \in \mathbb{R}^{M \times D}$ $(M = 64, D = 768)$ denote a learnable latent query tensor. Let visual patch representations be $\mathbf{X} \in \mathbb{R}^{N \times D}$ $(N = 1,024)$. Under $H = 8$ attention heads $(D_h = 64)$, keys are modulated via 2D-RoPE:

$$\mathbf{Q} = \mathrm{LayerNorm}(\mathbf{Z})\mathbf{W}_q, \quad \mathbf{K} = \mathbf{R}_{2D}(\mathrm{LayerNorm}(\mathbf{X})\mathbf{W}_k), \quad \mathbf{V} = \mathrm{LayerNorm}(\mathbf{X})\mathbf{W}_v$$

Cross-attention affinity $\mathbf{A} \in \mathbb{R}^{H \times M \times N}$ is evaluated as:

$$\mathbf{A}_{h,i,j} = \frac{\exp\left(\frac{1}{\sqrt{D_h}}\mathbf{Q}_{h,i}^{\top}\mathbf{K}_{h,j}\right)}{\sum_{l=1}^{N}\exp\left(\frac{1}{\sqrt{D_h}}\mathbf{Q}_{h,i}^{\top}\mathbf{K}_{h,l}\right)}$$

Theorem 2 (Computational Complexity Reduction). *Standard self-attention across $N = 1,024$ spatial patches incurs quadratic time complexity $\mathcal{O}(N^2 D) \approx 8.05 \times 10^8$ operations. The Perceiver Resampler reduces complexity to linear cross-attention $\mathcal{O}(MND) \approx 5.03 \times 10^7$ operations:*

$$\text{Compression Factor} = \frac{N^2 D}{MND} = \frac{N}{M} = \frac{1,024}{64} = 16.0\times$$

CHAPTER 12

Multi-Vector Late-Interaction (MaxSim) Operator

12.1 Mathematical Formulation

Let normalized query token vectors be $\mathbf{E}_q = (\mathbf{q}_1, \ldots, \mathbf{q}_{L_q})^\top \in \mathbb{R}^{L_q \times D}$ and document latents be $\mathbf{E}_d = (\mathbf{d}_1, \ldots, \mathbf{d}_M)^\top \in \mathbb{R}^{M \times D}$. The pairwise inner-product matrix is $\mathbf{S}_{i,j} = \mathbf{q}_i^\top \mathbf{d}_j$.

The late-interaction MaxSim score sums the maximum alignment of each query token across all document latents:

$$\mathrm{MaxSim}(\mathbf{E}_q, \mathbf{E}_d) = \sum_{i=1}^{L_q} \max_{j \in \{1, \ldots, M\}} \left(\mathbf{q}_i^\top \mathbf{d}_j\right)$$

With binary query attention mask $\mathbf{m}_q \in \{0, 1\}^{L_q}$, the masked score is normalized:

$$\mathrm{MaxSim}_{\mathrm{masked}}(\mathbf{E}_q, \mathbf{E}_d) = \frac{\sum_{i=1}^{L_q} \mathbf{m}_{q,i} \cdot \max_j (\mathbf{q}_i^\top \mathbf{d}_j)}{\sum_{i=1}^{L_q} \mathbf{m}_{q,i}}$$

CHAPTER 13

Symmetric Patch-InfoNCE Contrastive Criterion

13.1 Bidirectional Cross-Entropy

For a training batch of B document-query pairs $\{(\mathbf{Q}_b, \mathbf{D}_b)\}_{b=1}^{B}$ with late-interaction score matrix $\mathbf{M}_{b,c} = \mathrm{MaxSim}(\mathbf{Q}_b, \mathbf{D}_c)$ and temperature $\tau > 0$:

$$\mathcal{L}_{q \to d} = -\frac{1}{B} \sum_{b=1}^{B} \log \frac{\exp(\mathbf{M}_{b,b}/\tau)}{\sum_{j=1}^{B} \exp(\mathbf{M}_{b,j}/\tau)}$$

$$\mathcal{L}_{d \to q} = -\frac{1}{B} \sum_{c=1}^{B} \log \frac{\exp(\mathbf{M}_{c,c}/\tau)}{\sum_{i=1}^{B} \exp(\mathbf{M}_{i,c}/\tau)}$$

$$\mathcal{L}_{\text{Patch-InfoNCE}} = \frac{1}{2}(\mathcal{L}_{q \to d} + \mathcal{L}_{d \to q})$$

13.2 Learnable Inverse Temperature Dynamics

To ensure numerical stability, the temperature is parameterized in log-space $\theta_{\text{inv\_tau}} \in \mathbb{R}$ such that $\tau = \exp(-\theta_{\text{inv\_tau}})$. The analytical gradient with respect to parameter $\theta_{\text{inv\_tau}}$ is:

$$\frac{\partial \mathcal{L}_{q \to d}}{\partial \theta_{\text{inv\_tau}}} = \frac{\exp(\theta_{\text{inv\_tau}})}{B} \sum_{b=1}^{B} \left(\sum_{j=1}^{B} P(d_j \mid q_b) \mathbf{M}_{b,j} - \mathbf{M}_{b,b} \right)$$

As the model achieves cross-modal convergence and probability mass concentrates on positive pairs ($P(d_b \mid q_b) \to 1$), the gradient smoothly vanishes:

$$\lim_{P(d_b \mid q_b) \to 1} \frac{\partial \mathcal{L}}{\partial \theta_{\text{inv\_tau}}} = 0$$

CHAPTER 14

Parameter-Efficient Fine-Tuning (LoRA)

14.1 Low-Rank Decomposition

In Stage 2 generative reading, pre-trained projection weights $\mathbf{W}_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ are adapted via low-rank decomposition:

$$\mathbf{W} = \mathbf{W}_0 + \Delta \mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r} \mathbf{B} \mathbf{A}, \quad \mathbf{B} \in \mathbb{R}^{d_{\text{out}} \times r}, \ \mathbf{A} \in \mathbb{R}^{r \times d_{\text{in}}}$$

With rank $r = 16$ and scaling constant $\alpha = 32$, adapters modify $< 0.28\%$ of total model parameters while maintaining exact zero initialization ($\mathbf{B} = \mathbf{0} \implies \Delta \mathbf{W} = \mathbf{0}$ at step 0).

CHAPTER 15

Information Retrieval Metrics & Null-Hypothesis Expectations

15.1 Formal Metric Definitions

For evaluation query set $\mathcal{Q}$ and ground-truth document rank $\mathrm{rank}(q)$:

$$\mathrm{Recall}\,@\mathrm{K} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbf{1}[\mathrm{rank}(q) \leq K], \quad \mathrm{MRR} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{\mathrm{rank}(q)}$$

Theorem 3 (Expected MRR Under Random Null Hypothesis). *For an unconstrained candidate collection of size N, a uniform random permutation yields an expected Mean Reciprocal Rank defined by the normalized harmonic number:*

$$\mathbb{E}[\mathrm{MRR}_{\mathrm{random}}] = \frac{1}{N} \sum_{k=1}^{N} \frac{1}{k} = \frac{H_N}{N} \approx \frac{\ln N + \gamma}{N}$$

where $\gamma \approx 0.57721$ is the Euler-Mascheroni constant.

For our evaluation benchmark of $N = 1,000$ documents:

$$\mathbb{E}[\mathrm{MRR}_{\mathrm{random}}] \approx \frac{\ln(1000) + 0.57721}{1000} \approx \frac{6.9077 + 0.5772}{1000} = 0.007485 \approx 0.0075$$

Our experimental progression confirms statistically significant representation learning ($p < 10^{-12}$):

- Random Uniform Baseline: $\mathrm{MRR} = 0.0074$
- Stage 1 Baseline (186 steps): $\mathrm{MRR} = 0.0172$ ($2.3\times$ random)
- Scaled 10-Epoch (310 steps): $\mathrm{MRR} = 0.0302$ ($4.1\times$ random)
- **OmniDoc-RAG Final (12,330 steps):** $\mathrm{MRR} = \mathbf{0.1746}$ ($23.6\times$ **random /** $10.2\times$ **Stage 1**)